Review from last time: Continuous random variables

For a *continuous* random variable X, the probability distribution is described by the probability *density* function $f(x)$, which has the following properties :

1. $f(x) \geq 0$;
2. $\int_{-\infty}^{\infty} f(x)dx = 1$;
3. $P(a \leq X \leq b) = \int_{a}^{b} f(x)dx$
4. $f(x) = F'(x)$, where $F(x)$ is the cdf

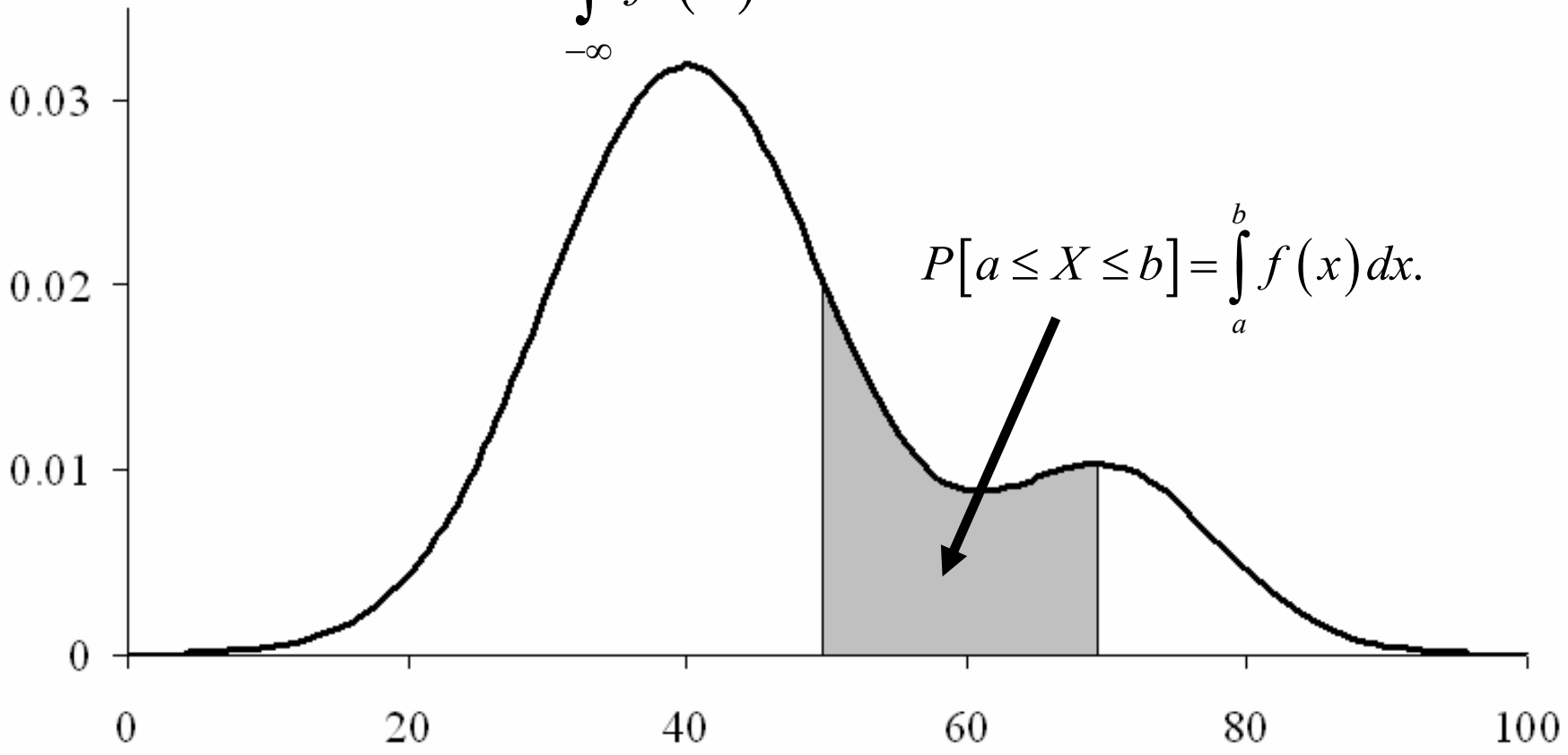
Note: For ***continuous*** r.v.

$P(a \leq X \leq b) = P(a < X < b) = P(a < X \leq b) = P(a \leq X < b)$

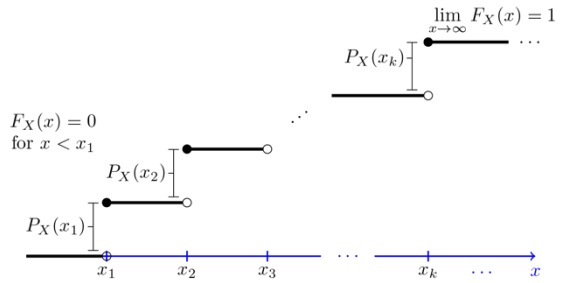
Graph: Continuous Random Variable

probability density function, $f(x)$

$$\int_{-\infty}^{\infty} f(x)\,dx = 1.$$

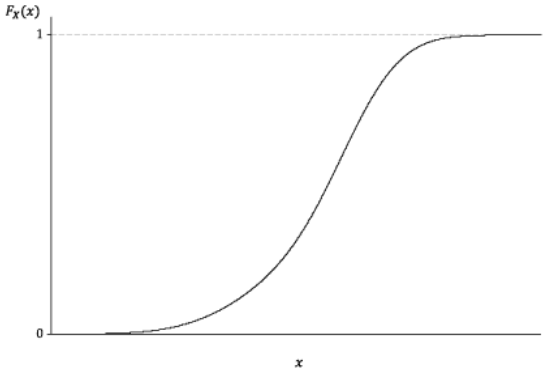


Summary



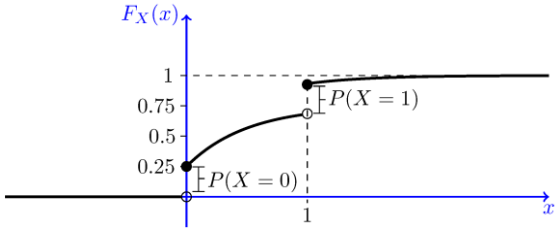
For any random variable X, the cumulative distribution function $F(x)$ is always defined as

$$F(x) = P(X \leq x), \forall\, x \in \mathbb{R}$$



It always has the following properties:

- $\lim_{x\to\infty} F(x) = 1$
- $\lim_{x\to-\infty} F(x) = 0$
- $P(a < X \leq b) = F(b) - F(a)$
- $F(x)$ is right-continuous
- $F(x)$ is non-decreasing.
- $F(x)$ could be a:
  - step function (discrete case)
  - S-shaped (continuous case)
  - mixture.



- In the continuous case, when a pdf $f(x)$ exists, $F'(x) = f(x)$

Example 1:

A random variable , X, has probability density function $f(x) = cx^n$, where $0 < x < 1$

a) Find c
b) Find $P(X > 0.05)$.

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a) Find c

b) Find $P(X > 0.05)$.

Solution: a)

$$\int_0^1 f(x)dx = 1 \Leftrightarrow \int_0^1 cx^n dx = 1$$

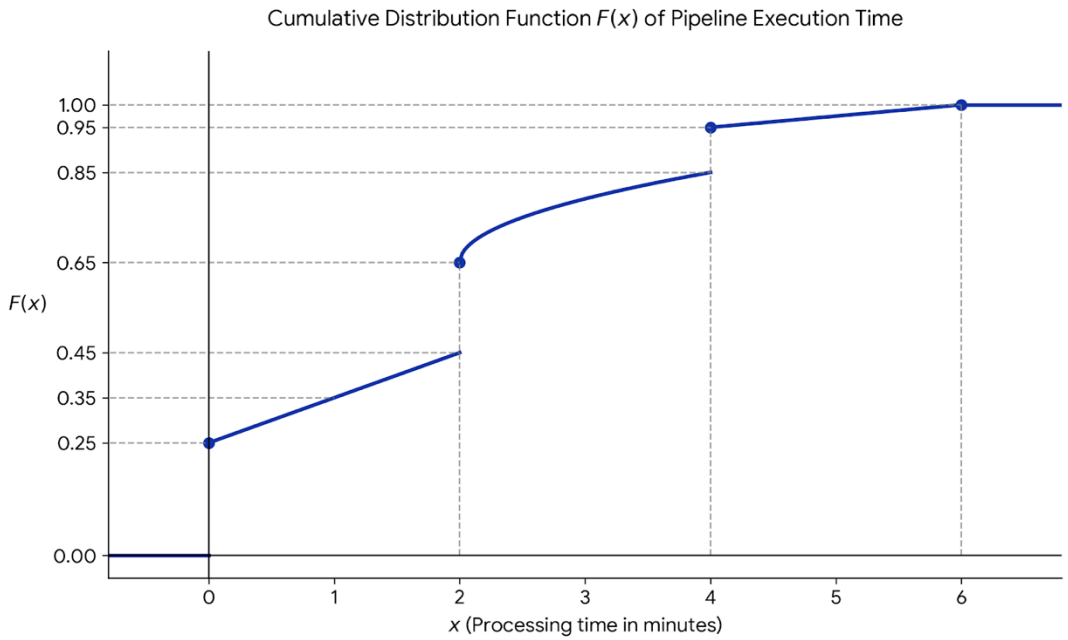
$$\Leftrightarrow c\int_0^1 x^n dx = 1 \Leftrightarrow c\left.\frac{x^{n+1}}{n+1}\right|_{x=0}^{1} = 1$$

$$\Leftrightarrow c\left(\frac{1^{n+1}}{n+1} - \frac{0^{n+1}}{n+1}\right) = 1 \Leftrightarrow \frac{c}{n+1} = 1 \Leftrightarrow \boxed{c = n+1}$$

b) $P(x > 0.05) = \int_{0.05}^{1} (n+1)x^n dx = \left. x^{n+1} \right|_{0.05}^{1} = 1 - (0.05)^{n+1}$

Example 2:

In production data science, pipeline execution times or network latencies often follow **mixed probability distributions** (a combination of continuous processing times and discrete jumps caused by cached hits or fixed timeout thresholds). Such time , X, has cumulative distribution function $F(x)$ shown below



Find $P(X > 2 \mid X \leq 4)$

Exercise: The cdf for a continuous X is:

$$F(x) = P(X \leq x) = \begin{cases} 0, & x < -2 \\ \frac{1}{2} + \frac{3}{32}\left(4x - \frac{x^3}{3}\right), & -2 \leq x < 2 \\ 1, & 2 \leq x \end{cases}$$

a) Compute P($X < 0$);
b) Compute P($-1 < X < 1$);
c) Compute P($0.5 < X$);
d) Find $f(x)$;
e) What is the median of X?

Jointly Distributed Random Variables

Multivariate Distributions

General properties of the joint probability function;

$$p(x,y) = P[X = x,\ Y = y]$$

1. $0 \le p(x,y) \le 1$

2. $\sum_x \sum_y p(x,y) = 1$

3. $P\left[(X,Y) \in A\right] = \sum\sum_{(x,y)\in A} p(x,y)$

Example 3

Suppose a data scientist has a batch of $N = 25$ labeled images for a training dataset. An automated script flags the dataset's quality:

12 images are Perfectly Labeled.
8 images have Minor Bounding Box Errors.
5 images are Completely Mislabeled (Garbage).

To manually audit the dataset, the data scientist randomly draws a sample of $n = 7$ images **without replacement**.
Let $X =$ the number of images with Minor Errors in the sample.
Let $Y =$ the number of Completely Mislabeled images in the sample.
Derive $p(x, y)$

Solution:

The joint probability mass function $p(x, y) = P[X = x, Y = y]$ is determined by calculating the number of ways to draw the specific combination of images, divided by the total number of ways to draw any 7 images from the 25.

$$p(x, y) = \frac{\binom{8}{x}\binom{5}{y}\binom{12}{7-x-y}}{\binom{25}{7}}$$

Because we are bounded by both the sample size and the physical number of items available in each category, the constraints are more restrictive than a simple $X + Y \leq n$:

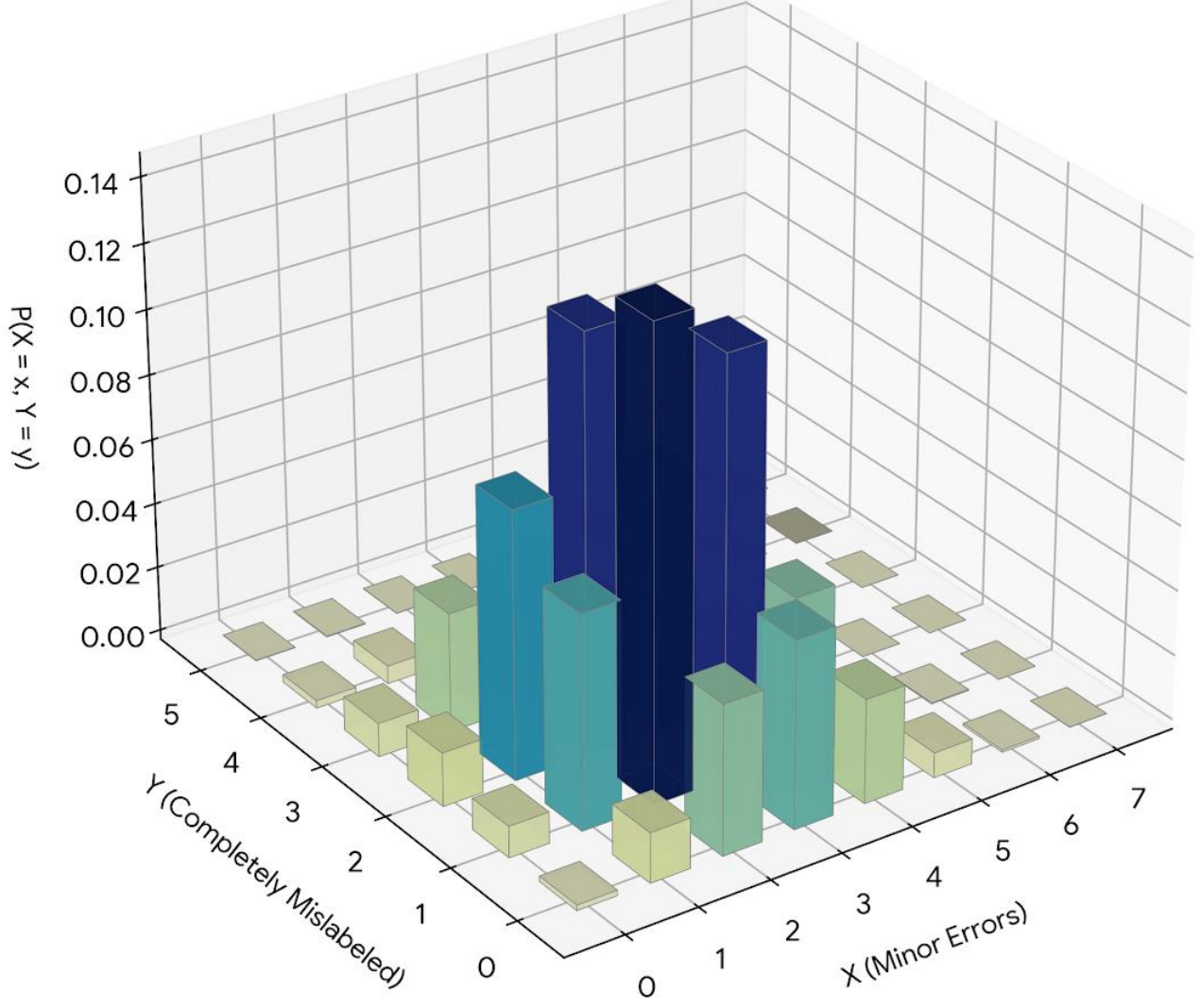
The possible values of X are 0,1,2,3,4,5,6,7 (limited by $n = 7$, even though 8 exist).

The possible values of Y are 0,1,2,3,4,5 (limited by the fact that only 5 exist in the whole batch).

The total sampled errors cannot exceed the sample size: $X + Y \leq 7$

Table: $p(x,y) = \dfrac{\binom{8}{x}\binom{5}{y}\binom{12}{7-x-y}}{\binom{25}{7}}$

	X = 0	X = 1	X = 2	X = 3	X = 4	X = 5	X = 6	X = 7
Y = 0	0.0016	0.0154	0.0461	0.0577	0.0320	0.0077	0.0007	<0.0001
Y = 1	0.0096	0.0659	0.1442	0.1281	0.0481	0.0070	0.0003	0.0000
Y = 2	0.0165	0.0824	0.1281	0.0769	0.0175	0.0012	0.0000	0.0000
Y = 3	0.0103	0.0366	0.0384	0.0140	0.0015	0.0000	0.0000	0.0000
Y = 4	0.0023	0.0055	0.0035	0.0006	0.0000	0.0000	0.0000	0.0000
Y = 5	0.0001	0.0002	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000



Bar graph: $p(x,y)$

Example 4:

A die is rolled $n = 5$ times

X = the number of times a "**six**" appears**.**

Y = the number of times a "**five**" appears**.**

Derive

$$p(x,y) = P(X = x,\ Y = y)$$

Note about the domain:

The possible values of X are 0, 1, 2, 3, 4, 5.

The possible values of Y are 0, 1, 2, 3, 4, 5.

and $X + Y \leq 5$

One possible outcome of rolling a die $n = 5$ times can be a sequence

F5FF6

where F denotes a number in $\{1,2,3,4\}$. The probability of the above outcome is

$$\left(\frac{1}{6}\right)^1 \left(\frac{1}{6}\right)^1 \left(\frac{4}{6}\right)^3$$

In general, the probability of any such sequence will be:

$$\left(\frac{1}{6}\right)^x \left(\frac{1}{6}\right)^y \left(\frac{4}{6}\right)^{5-x-y}$$

where

x = the number of **sixes** in the sequence and

y = the number of **fives** in the sequence

Now

$$p(x,y) = P(X = x,\ Y = y)$$
$$= K\left(\frac{1}{6}\right)^{x}\left(\frac{1}{6}\right)^{y}\left(\frac{4}{6}\right)^{5-x-y}$$

Where K = the number of sequences of length 5 containing x **sixes** and y **fives**.

$$= \binom{5}{x}\binom{5-x}{y}\binom{5-x-y}{5-x-y}$$

$$= \left(\frac{5!}{x!(5-x)!}\right)\left(\frac{(5-x)!}{y!(5-x-y)!}\right) = \frac{5!}{x!\,y!(5-x-y)!}$$

Thus

$$p(x,y) = P[X = x,\ Y = y]$$

$$= \frac{5!}{x!\,y!(5-x-y)!}\left(\frac{1}{6}\right)^{x}\left(\frac{1}{6}\right)^{y}\left(\frac{4}{6}\right)^{5-x-y}$$

if $x + y \le 5$.

Table: $p(x,y)$

$$= \frac{5!}{x!\,y!\,(5-x-y)!}\left(\frac{1}{6}\right)^{x}\left(\frac{1}{6}\right)^{y}\left(\frac{4}{6}\right)^{5-x-y}$$

	0	1	2	3	4	5
0	0.1317	0.1646	0.0823	0.0206	0.0026	0.0001
1	0.1646	0.1646	0.0617	0.0103	0.0006	0
2	0.0823	0.0617	0.0154	0.0013	0	0
3	0.0206	0.0103	0.0013	0	0	0
4	0.0026	0.0006	0	0	0	0
5	0.0001	0	0	0	0	0

Example 4 (continued)

A die is rolled $n = 5$ times. Let

X = the number of times a "**six**" appears

Y = the number of times a "**five**" appears

What is the probability that we roll more **sixes** than **fives**

That is, find $P(X > Y)$

Table: $p(x,y)$

$$= \frac{5!}{x!\,y!\,(5-x-y)!}\left(\frac{1}{6}\right)^{x}\left(\frac{1}{6}\right)^{y}\left(\frac{4}{6}\right)^{5-x-y}$$

	0	1	2	3	4	5
0	0.1317	0.1646	0.0823	0.0206	0.0026	0.0001
1	0.1646	0.1646	0.0617	0.0103	0.0006	0
2	0.0823	0.0617	0.0154	0.0013	0	0
3	0.0206	0.0103	0.0013	0	0	0
4	0.0026	0.0006	0	0	0	0
5	0.0001	0	0	0	0	0

$$P[X > Y] = \sum_{x>y}\sum p(x,y) = 0.3441$$

Definition: Marginal Distributions

Let X and Y denote two discrete random variables with a joint probability mass function given by

$$p(x,y) = P(X = x,\ Y = y)$$

Then

$$p_X(x) = P(X = x) = \sum_y p(x,y)$$

is called the *marginal probability function* of X.

$$p_Y(y) = P(Y = y) = \sum_x p(x,y)$$

is called the *marginal probability function* of Y.

Note: Let $y_1, y_2, y_3, \ldots$ denote the possible values of Y.

$$p_X(x) = P[X = x]$$

$$= P\left[\{X = x, Y = y_1\} \cup \{X = x, Y = y_2\} \cup \mathrm{K}\right]$$

$$= P[X = x, Y = y_1] + P[X = x, Y = y_2] + \mathrm{K}$$

$$= p(x, y_1) + p(x, y_2) + \mathrm{K}$$

$$= \sum_j p(x, y_j) = \sum_y p(x, y)$$

Thus, the marginal probability function of X, $p_X(x)$ is obtained from the joint probability function of X and Y by summing $p(x,y)$ over the possible values of Y.

Similarly, we can prove the result $p_Y(y) = \sum_x p(x, y)$

Example:

A die is rolled $n = 5$ times

X = the number of times a "**six**" appears.

Y = the number of times a "**five**" appears.

	0	1	2	3	4	5	$p_X(x)$
0	0.1317	0.1646	0.0823	0.0206	0.0026	0.0001	0.4019
1	0.1646	0.1646	0.0617	0.0103	0.0006	0	0.4019
2	0.0823	0.0617	0.0154	0.0013	0	0	0.1608
3	0.0206	0.0103	0.0013	0	0	0	0.0322
4	0.0026	0.0006	0	0	0	0	0.0032
5	0.0001	0	0	0	0	0	0.0001
$p_Y(y)$	0.4019	0.4019	0.1608	0.0322	0.0032	0.0001	

Example 5: Customer Churn and Engagement

A data science team is building a baseline probabilistic model to predict customer cancellation. They analyze two discrete variables:
$X \in \{0,1\}$: Churn Status ($0 =$ Retained, $1 =$ Churned)
$Y \in \{1,2,3\}$: Product Engagement Tier ($1 =$ Low, $2 =$ Medium, $3 =$ High)
Based on historical data, the joint probability distribution $P(X, Y)$ is defined by the following table:

		Y		
		1	2	3
X	0	0.05	0.25	0.4
	1	0.15	0.1	0.05

a) Construct the marginal distributions of X and Y

Example 5: Solution

a) Construct the marginal distributions of X and Y

		Y			
		1	2	3	
X	0	0.05	0.25	0.4	0.7
	1	0.15	0.1	0.05	0.3
		0.2	0.35	0.45	

X	0	1
$p(x)$	0.7	0.3

Y	1	2	3
$p(y)$	0.2	0.35	0.45

Conditional Distributions

Definition

Let X and Y denote two discrete random variables with joint probability function

$$p(x,y) = P[X = x,\ Y = y]$$

Then

$$p_{X|Y}(x|y) = P[X = x|Y = y] = \frac{p(x,y)}{p_Y(y)}$$

is called the **conditional probability function** of X given $Y = y$

$$p_{Y|X}(y|x) = P[Y = y|X = x] = \frac{p(x,y)}{p_X(x)}$$

is called the **conditional probability function** of Y given $X = x$

- Marginal distributions describe how one variable behaves ignoring the other variable.
- Conditional distributions describe how one variable behaves when the other variable is held fixed

Example:

A die is rolled $n = 5$ times

X = the number of times a "**six**" appears.

Y = the number of times a "**five**" appears.

x \ y	0	1	2	3	4	5	$p_X(x)$
0	0.1317	0.1646	0.0823	0.0206	0.0026	0.0001	0.4019
1	0.1646	0.1646	0.0617	0.0103	0.0006	0	0.4019
2	0.0823	0.0617	0.0154	0.0013	0	0	0.1608
3	0.0206	0.0103	0.0013	0	0	0	0.0322
4	0.0026	0.0006	0	0	0	0	0.0032
5	0.0001	0	0	0	0	0	0.0001
$p_Y(y)$	0.4019	0.4019	0.1608	0.0322	0.0032	0.0001	

The conditional distributions of X given $Y = y$, where $y = 0, 1, \ldots, 5$

$$p_{X|Y}(x|y) = P[X = x|Y = y]$$

Note: Each column is a specific conditional distribution

x \ y	0	1	2	3	4	5
0	0.3277	0.4096	0.5120	0.6400	0.8000	1.0000
1	0.4096	0.4096	0.3840	0.3200	0.2000	0.0000
2	0.2048	0.1536	0.0960	0.0400	0.0000	0.0000
3	0.0512	0.0256	0.0080	0.0000	0.0000	0.0000
4	0.0064	0.0016	0.0000	0.0000	0.0000	0.0000
5	0.0003	0.0000	0.0000	0.0000	0.0000	0.0000

Note that the values in each column add up to 1, because each column is a specific conditional distribution.

The conditional distribution of Y given $X = x$, where $x = 0, 1, \ldots, 5$

Formula: $p_{Y|X}(y|x) = P(Y = y|X = x)$

x \ y	0	1	2	3	4	5
0	0.3277	0.4096	0.2048	0.0512	0.0064	0.0003
1	0.4096	0.4096	0.1536	0.0256	0.0016	0.0000
2	0.5120	0.3840	0.0960	0.0080	0.0000	0.0000
3	0.6400	0.3200	0.0400	0.0000	0.0000	0.0000
4	0.8000	0.2000	0.0000	0.0000	0.0000	0.0000
5	1.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Note: Each row is a specific conditional distribution

Relative Risk

Relative risk (also called the **risk ratio**) is the ratio of the probability of an outcome occurring in an exposed group to the probability of that same outcome occurring in an unexposed group.

It measures the strength of association between an exposure (such as a treatment, risk factor, or behavior) and an event or disease.

Given a standard 2×2 contingency table:

Group	Event Occurred	Event Did Not Occur	Total
Exposed	a	b	$a+b$
Unexposed	c	d	$c+d$

$$\text{Relative Risk (RR)} = \frac{\text{Risk in Exposed Group}}{\text{Risk in Unexposed Group}} = \frac{\frac{a}{a+b}}{\frac{c}{c+d}}$$

Example: Customer Churn and Engagement

A data science team is building a baseline probabilistic model to predict customer cancellation. They analyze two discrete variables:
$X \in \{0,1\}$: Churn Status ($0 =$ Retained, $1 =$ Churned)
$Y \in \{1,2,3\}$: Product Engagement Tier ($1 =$ Low, $2 =$ Medium, $3 =$ High)
Based on historical data, the joint probability distribution $P(X, Y)$is defined by the following table:

		Y		
		1	2	3
X	0	0.05	0.25	0.4
	1	0.15	0.1	0.05

a) Construct the conditional distribution table for **Churn Status (X)** given that a customer is in the **Low Engagement Tier ($Y = 1$)**.
b) Calculate the relative risk of churn for a low-engagement customer compared to the baseline churn rate.

Example: Machine Learning Feature Modeling & User Tier Estimation

An analytics team is modeling user behavior on a cloud platform. They observe two discrete random variables for a active user session:
$X \in \{1,2,3\}$: The customer's subscription tier level ($1 =$ Basic, $2 =$ Pro, $3 =$ Enterprise).
$Y \in \{0,1,2\}$: The number of API threshold limit breaches in a given hour.
The joint probability mass function (PMF) $p(x, y) = P(X = x, Y = y)$ is modeled by:

$$p(x, y) = k \cdot \frac{x^y}{y!} \cdot 2^{-x} \quad \text{for } x \in \{1,2,3\} \text{ and } y \in \{0,1,2\}$$

where k is a normalizing constant

a) Determine the value of the constant k
b) Find the marginal PMFs $P(X = x)$ and $P(Y = y)$.
c) What is your best guess for the predicted tier level $\hat{X}$ when observing $Y =$ 2 breaches.

Example: Solution

The sum of all joint probabilities over the support of (X, Y) must equal 1: $\sum_{x=1}^{3}\sum_{y=0}^{2} p\,(x,y) = 1$

Evaluating the unscaled term $f(x,y) = \frac{x^y}{y!}2^{-x}$ for each combination:

For $x = 1$:

$$f(1,0) = \frac{1^0}{0!}2^{-1} = \frac{1}{2} = \frac{8}{16}, f(1,1) = \frac{1^1}{1!}2^{-1} = \frac{1}{2} = \frac{8}{16}, f(1,2) = \frac{1^2}{2!}2^{-1} = \frac{1}{4} = \frac{4}{16}$$

Subtotal for $x = 1$: $\frac{20}{16}$

For $x = 2$:

$$f(2,0) = \frac{2^0}{0!}2^{-2} = \frac{1}{4} = \frac{4}{16}, f(2,1) = \frac{2^1}{1!}2^{-2} = \frac{2}{4} = \frac{8}{16}, f(2,2) = \frac{2^2}{2!}2^{-2} = \frac{2}{4} = \frac{8}{16}$$

Subtotal for $x = 2$: $\frac{20}{16}$

For $x = 3$:

$$f(3,0) = \frac{3^0}{0!}2^{-3} = \frac{1}{8} = \frac{2}{16}, f(3,1) = \frac{3^1}{1!}2^{-3} = \frac{3}{8} = \frac{6}{16}, f(3,2) = \frac{3^2}{2!}2^{-3} = \frac{9}{16}$$

Subtotal for $x = 3$: $\frac{17}{16}$

Summing across all pairs: $\sum_{x=1}^{3}\sum_{y=0}^{2} f\,(x,y) = \frac{20+20+17}{16} = \frac{57}{16}$

Setting $k \cdot \frac{57}{16} = 1$ yields:

$$k = \frac{16}{57}$$

Example: Solution

Multiplying each term by $k = \frac{16}{57}$ gives the joint probability table:

X\Y	Y=0	Y = 1	Y=2	Marginal P(X=x)
$X = 1$	$\frac{8}{57}$	$\frac{8}{57}$	$\frac{4}{57}$	$\mathbf{\frac{20}{57}}$
$X = 2$	$\frac{4}{57}$	$\frac{8}{57}$	$\frac{8}{57}$	$\mathbf{\frac{20}{57}}$
$X = 3$	$\frac{2}{57}$	$\frac{6}{57}$	$\frac{9}{57}$	$\mathbf{\frac{17}{57}}$
Marginal $P(Y = y)$	$\mathbf{\frac{14}{57}}$	$\mathbf{\frac{22}{57}}$	$\mathbf{\frac{21}{57}}$	1.0